

Fares Mohamed

Alexandria, Egypt | faresmohamed260@gmail.com | github.com/faresmohamed260
linkedin.com/in/fares-mohamed-b83454194 | **Robotics & Embedded AI Engineer**

Summary

Fresh Computer Science graduate ranked first in class (CGPA: 3.97/4.00), building robotics-aware software, intelligent control interfaces, embedded prototypes, computer-vision systems, and AI automation with Python, C++, Arduino, and ESP32.

Experience

Armstrong

2023 – 2025

AI & Automation Engineer / Technical Content Creator (Part-time), Alexandria, Egypt

- Built AI-assisted content pipelines for the Digital Egypt Cubs Initiative (DECI), supporting a national technology-education program under Egypt's Ministry of Communications and Information Technology.
- Automated presentations, guides, videos, and visual assets using n8n, custom GPTs, open-source models, scripts, and repeatable review workflows.
- Developed robotics, Arduino, circuits, AI, and programming curricula with classroom-ready demonstrations and technical media.

Kayfa

Aug 2025 – Dec 2025

AI Subject Matter Expert, Alexandria, Egypt

- Authored practical lessons, assignments, and Python notebooks on machine learning, neural networks, CNNs, clustering, and anomaly detection.

Engineering for Kids-Egypt

2023 – 2025

AI & Robotics Instructor (Seasonal), Alexandria, Egypt

- Taught students aged 11–16 robotics, Arduino, circuits, AI, Python, and C++ through project-based seasonal programs.
- Guided builds from circuit assembly and sensor integration to programming, debugging, testing, and live demonstrations.

Projects

DUM-E: ESP32 Robotic Arm Platform

Python, C++, ESP32, Streamlit, Robotics

github.com/faresmohamed260/DUM-E

- Built a robotic arm platform combining ESP32 firmware, an operator dashboard, and guarded execution workflows for calibration, manual control, and repeatable task sequences.
- Implemented controller mapping, sequence recording, and FK/IK-assisted motion planning for pick-and-place demonstrations.

VisionDeck Computer Vision Suite

Python, OpenCV, MediaPipe, YOLO, Streamlit

- Developed a modular real-time dashboard for face detection, hand-gesture recognition, and object detection using webcam and Android IP-camera feeds.

SAGA + ComfyUI Generative Media Platform

Python, LangGraph, FastAPI, Supabase, ComfyUI, Modal

github.com/faresmohamed260/saga

- Architected typed LangGraph workflows for narrative analysis, structured generation, hybrid retrieval, persistent jobs, and automated tests.
- Built 10 local ComfyUI graphs across 130 renders for reference-consistent characters, ControlNet-guided generation, layered assets, editing, and face restoration; deployed selected Z-Image workflows on Modal with caching, warmup, monitoring, and failover.

Education

Alexandria University, Faculty of Computer and Data Science

2022 – 2026

B.Sc. Computer Science (Intelligent Systems) | **Class Rank: 1st** | **CGPA: 3.97/4.00**

Skills

Robotics & Embedded: Arduino, ESP32, C++, sensors, circuits, controller mapping, calibration, FK/IK-assisted motion planning

AI & Vision: Python, OpenCV, MediaPipe, YOLO, LangGraph, RAG, ComfyUI, ControlNet, Z-Image, Qwen-Image, FLUX.2

Software & Platforms: FastAPI, React, Streamlit, Docker, Git, pytest, PostgreSQL, Supabase, Modal, n8n | **Languages:** Arabic, English

Leadership & Certifications

Head of AI Committee, ElCoder – Led workshops on AI and ML fundamentals. | Machine Learning Specialization | Deep Learning Specialization
| Google Cloud Big Data and ML Fundamentals